1. The Big Picture: What is Cost of Capital?

Imagine you're running a lemonade stand. To buy lemons and sugar, you need money. You could borrow it from your parents (debt) or use your own savings (equity). Both sources of money have a "cost." The cost of borrowing from your parents is the interest you pay them. The cost of using your own savings is the money you could have earned by putting it in a bank account instead.

In the world of a company, the **cost of capital** is the minimum return a company must earn on its projects to satisfy its investors (those who provide the money). If a company's investment doesn't make enough money to cover this cost, it's a bad investment because it destroys value for the owners.

Components of Cost of Capital

Cost of Debt (Kd)

The **cost of debt** is the effective interest rate a company pays on its borrowings, such as loans or bonds.[5] A key feature of debt is that the interest payments are **tax-deductible**.[6] This creates a "tax shield" that reduces the company's overall tax burden.[7] Therefore, we always calculate the **after-tax cost of debt**.

- **Calculation:** The simplest way to calculate the cost of debt is to find the interest rate on the debt and adjust it for the tax shield.[8]
    - $K_d = \text{Interest Rate} \times (1 - \text{Tax Rate})$
- **Example:** If a company borrows money at a 10% interest rate and its corporate tax rate is 30%, the after-tax cost of debt is [9]$10\% \times (1 - 0.30) = 7\%$.[10] This means for every dollar of debt, the company's effective cost is only 7 cents.

Cost of Equity (Ke)

The **cost of equity** is the return required by a company's common shareholders for their investment.[11] It's the most difficult component to

calculate because unlike debt, there are no fixed, contractual payments. Shareholders are compensated through dividends and capital appreciation, and they are last in line to be paid if the company fails.[12] Because of this higher risk, the cost of equity is almost always **higher than the cost of debt**.[13] There are two common methods to estimate the cost of equity:[14]

- **Dividend Discount Model (DDM):** This model works best for companies that pay stable dividends. It assumes the value of a stock is the present value of all future dividends.[15]
    - **Formula:** Ke=P0D1+g
    - D1: Expected dividend next year.
    - P0: Current market price per share.
    - g: Expected dividend growth rate.[16]
- **Capital Asset Pricing Model (CAPM):** This model links an investment's required return to its risk relative to the overall market.[17]
    - **Formula:** [18]Ke=Rf+β×(Rm−Rf)
    - Rf: Risk-free rate of return (e.g., return on a government bond).[19]
    - β: Beta, a measure of the stock's volatility relative to the market.[20]
    - (Rm−Rf): Market risk premium.[21]

Cost of Preference Capital (Kp)

The **cost of preference capital** is the fixed dividend rate a company pays to its preferred shareholders.[22] Preference shares are a hybrid of debt and equity. Like debt, they have a fixed payment, but like equity, dividends are **not tax-deductible**. Preferred shareholders have a higher claim on a company's assets than common shareholders in the event of liquidation, but a lower claim than debt holders.

- **Calculation:** The cost is the fixed dividend payment divided by the market price of the preference share.
    - Kp=Market Price of Preference ShareFixed Dividend

The Weighted Average Cost of Capital (WACC)

The **Weighted Average Cost of Capital (WACC)** is the overall, blended cost of a company's financing, reflecting the proportion of each funding source in its capital structure.[23] It's the most widely used measure of a firm's cost of capital. A company's goal is often to minimize its WACC to maximize its value.[24]

- **Concept:** WACC is a weighted average of the costs of each individual funding source.[25] The "weights" are the proportion of each type of capital in the company's capital structure (the mix of debt, equity, and preference capital).[26]
- **Formula:**
  $WACC=(Wd\times Kd(1-T))+(Wp\times Kp)+(We\times Ke)$
  - Wd, Wp, We: The proportions of debt, preference capital, and equity in the firm's capital structure. These are calculated based on the market value of each component.[27]
  - Kd, Kp, Ke: The individual costs of debt, preference capital, and equity.[28]
- **Importance:** WACC is the discount rate used in capital budgeting decisions, like Net Present Value (NPV) analysis, to determine if a project is financially viable.[29] If a project's expected return is greater than the company's WACC, it's considered a value-creating investment.[30]

Corporate Finance Basics

1. Business Risk vs. Financial Risk

A. Business Risk

- Risk of not being able to cover operating costs.

- Comes from the **nature of business**, not from financing.

- Factors: demand & sales fluctuations, competition, input cost changes, fixed costs.

- Example: A cinema theatre has high business risk because sales depend on new movie releases.

Business Risk Example

- Company A sells only ice cream.

- Fixed operating cost = ₹50,000 per month.

- Summer sales = ₹1,50,000 (profit after costs = ₹1,00,000).

- Winter sales = ₹60,000 (profit after costs = only ₹10,000).

Fluctuation in profit because of seasonal demand = Business Risk.
(It exists whether the company borrows money or not).

B. Financial Risk

- Risk to shareholders due to using **debt financing (leverage)**.

- Debt requires **fixed interest payments**, regardless of profit.

- More debt = higher financial risk.

- Example: A company takes a bank loan → if profits fall, it may still struggle to pay interest.

Total Risk faced by shareholders = Business Risk + Financial Risk.

Financial Risk Example

- Suppose the same Company A takes a loan of ₹5,00,000 at 10% interest.
- Annual interest = ₹50,000.
- If profit before interest = ₹80,000 → Net profit = ₹30,000.
- If profit before interest = ₹40,000 → Net profit = –₹10,000 (loss).

Obligation to pay fixed interest causes extra risk = Financial Risk.

2. Capital Structure Theories

Capital structure = **mix of debt & equity** in financing.
These theories explain whether the mix affects **value of the firm** & **cost of capital**.

A. Net Income (NI) Approach

- Assumption: Cost of debt (Kd) < Cost of equity (Ke).
- If firm uses more debt → average cost of capital falls → firm value rises.
- **Implication**: Best to use as much debt as possible.
- **Criticism**: Too much debt increases risk in reality.
  Idea: Debt is cheaper → more debt lowers cost of capital.

Example:

1. **All Equity Firm**
    - Equity = ₹10,00,000
    - Net profit = ₹2,00,000
    - Ke = 15% → Equity value = Profit / Ke = 2,00,000 / 0.15 = ₹13,33,333
    - Firm Value = ₹13,33,333
2. **50% Debt, 50% Equity**
    - Debt = ₹5,00,000 @ 10% → Interest = ₹50,000
    - Profit after interest = ₹1,50,000
    - Ke = 15% → Equity value = 1,50,000 / 0.15 = ₹10,00,000
    - Firm Value = Equity + Debt = 10,00,000 + 5,00,000 = ₹15,00,000

Firm value **increased** when debt was used → supports NI theory.

B. Net Operating Income (NOI) Approach

- Assumption: Overall cost of capital (Ko) is **constant**, regardless of debt.
- More debt → cost of equity (Ke) rises (due to higher risk).

- Cheap debt benefit gets cancelled.
- **Implication**: Capital structure is irrelevant, firm value doesn't change.
- Idea: Overall cost of capital is constant. More debt → Ke increases, cancelling the benefit.

Example:

- Assume Ko (overall cost) = 12%.
- EBIT = ₹2,00,000.
- Firm Value = EBIT / Ko = 2,00,000 / 0.12 = ₹16,66,667 (constant).

Firm value does **not change** whether debt or equity is used.

C. Modigliani–Miller (MM) Approach

1. **Without Taxes (1958):**
    - Perfect markets assumed (no taxes, no bankruptcy costs).
    - Capital structure irrelevant → firm value depends only on profits from assets.
2. **With Taxes (1963):**
    - Interest is tax deductible → debt gives a **tax shield**.
    - Value of Levered Firm (VL) = Value of Unlevered Firm (VU) + (Tax rate × Debt).

- **Implication**: More debt → higher value of firm.

(i) Without Taxes

- Value of firm is same under any structure.
- Example: Firm Value = EBIT / Ko = 2,00,000 / 0.12 = ₹16,66,667.
- Same result whether debt = 0% or 50%.

Capital structure irrelevant.

(ii) With Taxes

- Debt gives a tax shield (Tax × Debt).

Example:

- Unlevered Firm Value (VU) = EBIT × (1 – Tax) / Ke = 2,00,000 × 0.70 / 0.15 = ₹9,33,333
- If firm takes debt of ₹5,00,000:
  VL = VU + (Tax × Debt) = 9,33,333 + (0.30 × 5,00,000) = ₹10,83,333

Value rises when debt is used → supports MM with taxes.

D. Traditional Approach (Intermediate)

- A compromise view.

- At first, using debt reduces overall cost of capital → value increases.
- Beyond a point, too much debt raises cost of equity heavily → overall cost rises.
- **Implication**: There exists an **Optimal Capital Structure** (best debt-equity mix).
- Idea: Some debt is good, too much is bad.

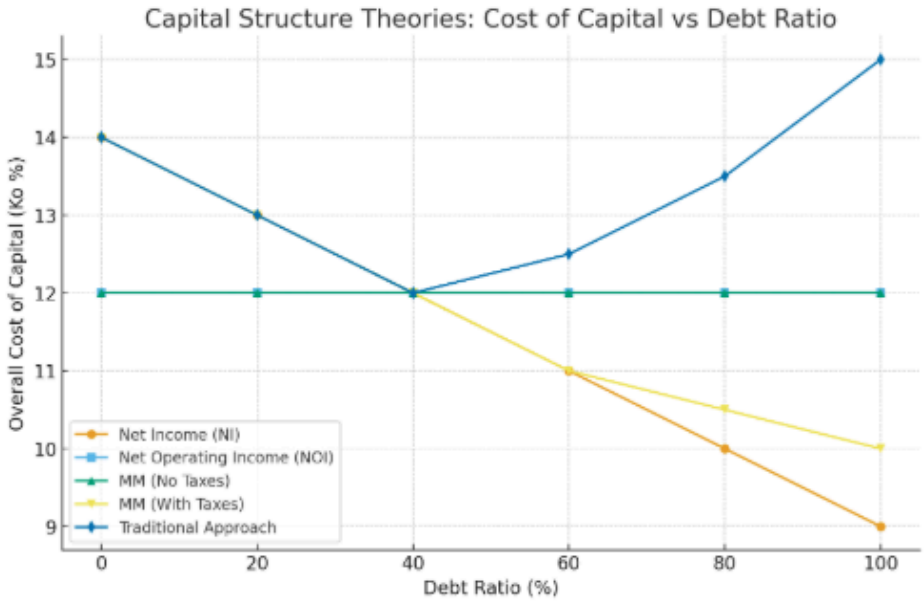
Example:

1. **0% Debt** → Ko = 14% → Value = 2,00,000 / 0.14 = ₹14,28,571
2. **30% Debt** → Ko = 12% → Value = 2,00,000 / 0.12 = ₹16,66,667
3. **70% Debt** → Ko = 15% → Value = 2,00,000 / 0.15 = ₹13,33,333

Value first increases with debt (benefit), then decreases (risk) → there is an **optimal mix**.

✅ Final Recap Table (with Example Insights)

Theory	Effect of Debt	Example Result
NI	More debt ↑ value	Value rose from ₹13.3L to ₹15L
NOI	No effect	Value constant at ₹16.6L
MM (No Tax)	No effect	Value constant at ₹16.6L
MM (With Tax)	More debt ↑ value	Value rose from ₹9.3L to ₹10.8L
Traditional	Some debt ↑ value, too much ↓ value	Optimal at 30% debt (₹16.6L)



Here's a graph 📊 showing how the **overall cost of capital (Ko)** changes with increasing debt under each capital structure theory:

- **Net Income (NI):** Ko falls as debt rises.
- **Net Operating Income (NOI) & MM (No Taxes):** Ko stays constant.
- **MM (With Taxes):** Ko falls gradually due to tax shield.
- **Traditional Approach:** Ko first falls (benefit of debt), then rises (too much risk) → U-shaped curve.

1. Capital Structure and Its Importance

Capital structure refers to the way a firm finances its assets through a combination of **debt** (loans, bonds) and **equity** (share capital, retained earnings).

- **Debt** → Borrowed funds, interest is tax-deductible, fixed obligation.
- **Equity** → Ownership funds, dividends are not obligatory.

Objective of capital structure: Maximize **firm value** and **minimize cost of capital**.

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2. Theories of Capital Structure

2.1 Net Income (NI) Approach

- **Assumption**: There is a **direct relationship** between leverage and firm value.
- **Key Idea**: Using more debt increases the **Return on Equity (ROE)** because debt is cheaper than equity.

- **Implication**: More debt → Higher EPS → Higher firm value.

Example:

- EBIT = ₹1,00,000
- Debt = ₹2,00,000 @ 10% interest
- Equity = ₹3,00,000

Interest on debt = 10% of ₹2,00,000 = ₹20,000
Net Income = EBIT – Interest = 1,00,000 – 20,000 = ₹80,000

- EPS = Net Income ÷ Equity = 80,000 ÷ 3,00,000 ≈ 26.7%

If debt increases, interest rises slightly, but EPS increases because equity decreases.

Limitation: Ignores **financial risk**.

2.2 Net Operating Income (NOI) Approach

- **Assumption**: **Value of the firm is independent of capital structure**.
- **Key Idea**: Total value = Value of debt + Value of equity = Constant.
- **Cost of equity** rises with debt, but **overall cost of capital (WACC)** remains constant.

Graphically: EPS increases with leverage, but firm value stays **same**.

Example:

- EBIT = ₹1,00,000
- Equity = ₹5,00,000, Debt = ₹0 → Net income = ₹1,00,000
- If debt = ₹2,00,000, interest = ₹20,000 → Net income = ₹80,000, but risk increases.
- WACC remains constant → firm value unchanged.

2.3 Modigliani-Miller (MM) Approach

Developed by **Franco Modigliani & Merton Miller (1958)**.

Without Taxes

- **Proposition I**: Value of firm **V = constant**, independent of capital structure.
- **Proposition II**: Cost of equity (Ke) rises linearly with debt, but **overall WACC remains constant**.

With Taxes

- Interest is tax-deductible → **levered firm value > unlevered firm value**.
- **V(L) = V(U) + (Tc × D)**
    - Tc = corporate tax rate, D = debt

- Debt increases value due to **tax shield**.

Example:

- Unlevered firm value = ₹5,00,000
- Debt = ₹2,00,000, Tax rate = 30%
- Tax shield = 2,00,000 × 30% = ₹60,000
- Levered firm value = 5,00,000 + 60,000 = ₹5,60,000

2.4 Traditional Approach

- Combines **NI and NOI approaches**.
- **Key Idea**: Up to an optimum level of debt, **WACC decreases** and firm value increases.
- Beyond optimum, financial risk dominates → WACC increases.

Graph:

- X-axis = Debt ratio
- Y-axis = WACC & Firm value
- Shows a **U-shaped curve for WACC**, **inverted U-shaped curve for firm value**.

Example:

- Debt ratio 0–30% → firm value rises, WACC falls
- Debt ratio >30% → firm value falls, WACC rises

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3. Factors Influencing Capital Structure Decisions

1. **Business Risk** – Higher operational risk → lower debt to avoid financial distress.
2. **Financial Risk** – High debt increases risk → may limit borrowing.
3. **Tax Considerations** – Interest is tax-deductible → tax shield benefit.
4. **Flexibility** – Firm may keep debt capacity for future needs.
5. **Growth Rate** – High-growth firms rely more on equity to avoid interest burden.
6. **Control Considerations** – Issuing debt avoids dilution, equity dilutes ownership.
7. **Market Conditions** – Interest rates, investor sentiment affect debt/equity choice.

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4. EBIT-EPS Analysis

Purpose: Find the **optimal capital structure** by comparing **Earnings per Share (EPS)** under different financing options.

- **EBIT** = Earnings before interest and tax
- **EPS** = (EBIT – Interest) ÷ No. of shares

Steps:

1. Calculate EPS for different debt-equity combinations at different EBIT levels.
2. Compare EPS → choose the combination giving **highest EPS at expected EBIT**.

Example:

Option	Debt	Equity	Interest	EBIT	EPS
A	0	10,000	0	1,00,000	10
B	2,00,000	8,000	20,000	1,00,000	(1,00,000-20,000)/8,000 = 10

- Decision depends on **expected EBIT**; higher EBIT favors debt (financial leverage).

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5. Leverage in Corporate Finance

Leverage is the **use of debt** to increase returns.

Types of Leverage

1. **Operating Leverage** – Related to **fixed costs in operations**
    - Formula: Degree of Operating Leverage (DOL) = % change in EBIT ÷ % change in Sales
2. **Financial Leverage** – Related to **fixed financial costs (interest)**
    - Formula: Degree of Financial Leverage (DFL) = % change in EPS ÷ % change in EBIT
3. **Combined Leverage** – Combines operating + financial leverage
    - DCL = DOL × DFL

Example:

- EBIT rises by 10%, sales increase by 5%, EPS rises by 15% → shows effect of leverage.

Risk: Leverage magnifies **returns** but also **losses**.

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6. Summary Table of Theories

Theory	WACC	Firm Value	View on Debt
NI	Falls with debt	Increases with debt	More debt → higher EPS/value
NOI	Constant	Constant	Capital structure irrelevant
MM (no tax)	Constant	Constant	Irrelevant

MM (with tax)	Falls with debt	Increases with debt	Debt adds value via tax shield
Traditional	Falls then rises	Rises then falls	Optimal debt exists

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✅ Key Takeaways

- Optimal capital structure balances **risk & return**.
- Debt increases EPS but also financial risk.
- Taxes, business risk, and growth influence capital structure decisions.
- EBIT-EPS analysis helps **practically choose financing mix**.
- Leverage magnifies **profit & loss**, so careful management is essential.

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1. Factors Influencing Capital Structure Decisions

Capital structure refers to the **mix of debt and equity** used by a firm to finance its operations and growth.
Choosing the right combination affects the **risk, cost of capital, and overall value of the firm**.

A. Internal Factors

Factor	Explanation	Example

1. Nature of Business	Capital-intensive industries (steel, power) prefer more debt, while service firms (IT, consulting) rely more on equity.	Infosys (low debt) vs. Tata Steel (high debt)
2. Stability of Earnings	Firms with stable profits can safely use more debt (interest payments covered easily).	Power Grid Corporation (stable returns → more debt)
3. Cash Flow Position	Strong cash flows = high debt capacity; weak flows → prefer equity.	Reliance Industries' robust cash flow supports large borrowings.
4. Control Considerations	Issuing equity may dilute ownership; promoters prefer debt to retain control.	Founders of startups often prefer loans over VC funding.
5. Cost of Capital	Debt is cheaper (tax-deductible interest) but increases financial risk. Firms aim for **minimum weighted average cost of capital (WACC)**.	Companies compute WACC to find the optimum mix.
6. Flexibility	Firm should maintain borrowing capacity for future needs.	A firm may not exhaust all debt options now to keep future flexibility.
7. Risk of Insolvency	Higher debt → higher fixed obligations → greater bankruptcy risk.	Airlines avoid heavy leverage due to cyclic demand.

B. External Factors

Factor	Explanation	Example
1. Market Conditions	In bullish markets, equity is easier to issue; in recession, firms prefer debt.	Many Indian firms issued IPOs in 2021 bull run.
2. Tax Policy	Interest on debt is tax-deductible → encourages debt financing.	Corporate tax benefits make debt cheaper.
3. Lending Attitude of Financial Institutions	Availability of loans and interest rate policies affect decisions.	Banks' credit policies influence corporate debt levels.
4. Legal Regulations	SEBI, RBI, or Companies Act norms may limit debt ratios or public issues.	NBFCs follow RBI's capital adequacy norms.
5. Economic Conditions	In inflationary conditions, firms may prefer fixed-rate debt; in deflation, equity is safer.	FMCG firms adjust financing based on inflation trends.

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2. EBIT–EPS Analysis

Meaning:

EBIT–EPS (Earnings Before Interest and Taxes – Earnings Per Share) analysis helps in **comparing different financing plans** (all equity, debt, or preference) to determine which one gives **higher EPS** at a particular EBIT level.

Steps:

1. Estimate EBIT (operating profit).
2. Calculate **Interest** on debt under each plan.
3. Subtract interest → get **EBT (Earnings Before Tax)**.
4. Apply **Tax rate** → get **EAT (Earnings After Tax)**.
5. Divide by **Number of Equity Shares** → get **EPS**.

Formula:

$$\text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1 - \text{Tax Rate})}{\text{Number of Shares}}$$

Example:

A company needs ₹10,00,000 for expansion.

Plan	Debt	Equity	Interest @ 10%	No. of Shares (₹100 each)
Plan A	₹0	₹10,00,000	₹0	10,000
Plan B	₹5,00,000	₹5,00,000	₹50,000	5,000

Plan C	₹8,00,000	₹2,00,000	₹80,000	2,000

Assume EBIT = ₹2,00,000 and Tax = 40%.

Plan	EBIT	Interest	EBT	Tax (40%)	EAT	EPS (₹)
A	2,00,000	0	2,00,000	80,000	1,20,000	12
B	2,00,000	50,000	1,50,000	60,000	90,000	18
C	2,00,000	80,000	1,20,000	48,000	72,000	36

🔹 **Inference:**
Plan C gives the highest **EPS = ₹36**, meaning higher **financial leverage** benefits shareholders **if EBIT is sufficiently high**.

However, if EBIT falls below a certain level, **interest burden** makes EPS decline sharply — that level is called the **Indifference Point**.

Indifference Point Formula:

$$(EBIT - I1)(1 - t)/N1 = (EBIT - I2)(1 - t)/N2$$

Solving gives the EBIT where both plans give **equal EPS**.

⚙️ 3. Leverage Analysis

Leverage means using **fixed costs** (either operating or financial) to **magnify returns**.

A. Types of Leverage

Type	Meaning	Formula	Effect
Operating Leverage (DOL)	Effect of fixed operating costs on EBIT	% change in EBIT / % change in Sales	High DOL → higher business risk
Financial Leverage (DFL)	Effect of interest cost on EPS	% change in EPS / % change in EBIT	High DFL → higher financial risk
Combined Leverage (DCL)	Combined effect on EPS	DOL × DFL	Indicates total risk

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Example:

Particulars	Amount (₹)
Sales	10,00,000
Variable Cost	6,00,000
Fixed Cost	2,00,000
Interest	1,00,000

EBIT = 10,00,000 – 6,00,000 – 2,00,000 = 2,00,000

$$DOL = \frac{\text{Contribution}}{\text{EBIT}} = \frac{(10,00,000 - 6,00,000)}{2,00,000} = 2$$

$$DFL = \frac{\text{EBIT}}{\text{EBIT - Interest}} = \frac{2,00,000}{1,00,000} = 2$$

$$DCL = DOL \times DFL = 2 \times 2 = 4$$

🔹 Interpretation:
A 1% change in sales will cause a **4% change in EPS** — showing **high total risk and reward**.

🗒️ **Summary Table**

Concept	Meaning	Key Metric / Formula	Impact
Capital Structure	Mix of debt and equity	Debt–Equity Ratio	Affects cost of capital and risk
EBIT–EPS Analysis	Compare financing plans	EPS = (EBIT–Interest)(1–T)/Shares	Helps choose best plan
Leverage	Amplification of profits	DOL, DFL, DCL	Indicates business and financial risk

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